

User Guide (English)

Project: Download Bank of Taiwan Exchange Rate

Audience: Business / operations / NetSuite administrators (no coding required for basic setup)

Chinese version: USER_GUIDE.zh-TW.md (./USER_GUIDE.zh-TW.md)

Technical guide: TECHNICAL.en.md (./TECHNICAL.en.md)

1. What does this system do?

Bank of Taiwan (BOT) publishes daily foreign-exchange rates as a CSV file. The official site uses bot protection, so NetSuite usually cannot download it directly.

This project:

1. Downloads the BOT daily FX CSV automatically
2. Converts it to JSON
3. Publishes it on a free CDN (jsDelivr)
4. Lets NetSuite fetch a stable HTTPS URL and use rate fields immediately

BOT official CSV → GitHub auto-update → CDN → NetSuite downloads JSON

2. URLs you need

Production (recommended)

<https://cdn.jsdelivr.net/gh/ChesleyLo/DownloadBankOfTaiwanExchangeRate@main/data/bot-xrt-latest.json>

Fallback (if CDN is stale or unavailable)

<https://raw.githubusercontent.com/ChesleyLo/DownloadBankOfTaiwanExchangeRate/main/data/bot-xrt-latest.json>

Raw CSV (optional)

<https://cdn.jsdelivr.net/gh/ChesleyLo/DownloadBankOfTaiwanExchangeRate@main/data/bot-xrt-latest.csv>

Open the JSON URL in a browser. You should see something like:

```
{
  "source": "Bank of Taiwan",
  "base": "TWD",
  "byCurrency": {
    "USD": {
      "cash": { "buy": 31.87, "sell": 32.54 },
      "spot": { "buy": 32.195, "sell": 32.345 }
    }
  }
}
```

}

3. How to read rate fields

Field path	Meaning
byCurrency.USD.cash.buy	USD cash bid (bank buy)
byCurrency.USD.cash.sell	USD cash ask (bank sell)
byCurrency.USD.spot.buy	USD spot bid
byCurrency.USD.spot.sell	USD spot ask
byCurrency.USD.forward.buy.30	USD 30-day forward bid
byCurrency.USD.forward.sell.30	USD 30-day forward ask

- Base currency: base = TWD
- Currency codes: USD, EUR, JPY, CNY, ...
- Missing quotes are null (not 0)
- For NetSuite exchange-rate import, teams often use spot ask: spot.sell

The authoritative currency list is whatever appears in the day's rates / byCurrency.

4. How often is data refreshed?

Three layers (details: SCHEDULING.en.md (./SCHEDULING.en.md)):

Layer	Times (Taiwan, weekdays)
Primary cron-job.org	hourly 09:10–16:10 + 18:30
Backup GitHub cron	10:30, 15:30, 18:30
Local Mac (optional)	same; Mac must be on

Notes:

- No commit when rates are unchanged (workflow may still run)
- No weekend runs by default
- If scheduling seems stuck, see SCHEDULING.en.md §0.2 (./SCHEDULING.en.md#02-schedule-not-running-step-by-step)
- How to confirm runs see SCHEDULING.en.md §0.5 (./SCHEDULING.en.md#05-how-to-confirm-the-schedule-ran-day-to-day)
- Manual trigger: ./scripts/trigger_update.sh or Actions → Run workflow

5. NetSuite setup (administrator)

5.1 Prerequisites

- Permission to create Scripts / Deployments

- Permission to write to the File Cabinet
- Outbound HTTPS allowed to cdn.jsdelivr.net

5.2 Upload the script

1. Open netsuite/DownloadBotRates_SS2.js
2. In NetSuite: Customization → Scripting → Scripts → New
3. Upload the JS file as a Scheduled script
4. Save

5.3 Optional script parameters

Parameter ID	Type	Purpose
custscript_bot_cdn_json_url	Free-Form Text	Override CDN URL
custscript_bot_rate_field	Free-Form Text	e.g. spot.sell
custscript_bot_apply_rates	Checkbox	Enable rate-write logic (off by default; logs only)

If parameters are not created, the script uses the hardcoded default CDN URL.

5.4 Folder ID

FOLDER_ID defaults to -15 (File Cabinet root).

Change it to your target folder's internal ID.

5.5 Deployment

1. Create a Scheduled Script Deployment
2. Status: Released
3. Schedule: once daily after Taiwan market close (e.g. 17:00 or 19:00 local)
4. Run Execute Now once to validate

5.6 What success looks like

1. Downloads JSON from CDN
2. Parses with JSON.parse
3. Saves to File Cabinet:
 - bot-xrt-latest.json
 - bot-xrt-YYYY-MM-DD.json
1. Writes an audit log entry (including a USD spot-sell summary)

On GitHub, historical files live under data/history/ and only the last 90 days are kept. NetSuite File Cabinet naming is independent.

By default the script does not create NetSuite Currency Rate records. Enabling that requires a code change described in the technical guide.

6. Using rates in reports / processes

Typical pattern:

1. Scheduled Script stores the latest JSON in the File Cabinet daily
2. Another script/workflow reads the file and takes `byCurrency.<CCY>.spot.sell`
3. Writes to your rate table, custom record, or transaction conversion logic

Conceptual example:

```
const payload = JSON.parse(cdnResponseBody);  
const rate = payload.byCurrency.USD.spot.sell; // e.g. 32.345
```

7. FAQ

Q1. CDN says the file was not found?

The GitHub repository is probably private. jsDelivr only serves public repos. Confirm:

<https://github.com/ChesleyLo/DownloadBankOfTaiwanExchangeRate>

Q2. Rates look stale?

Possible causes:

1. Rates have not changed today (no commit)
2. jsDelivr cache not refreshed
3. Schedule did not trigger → SCHEDULING.en.md §0.2
(./SCHEDULING.en.md#02-schedule-not-running-step-by-step)

Q3. Schedule not running? / How to confirm it ran?

Day-to-day checks (fastest first):

1. GitHub Actions (<https://github.com/ChesleyLo/DownloadBankOfTaiwanExchangeRate/actions>) — look for successful workflow_dispatch
2. cron-job.org History (<https://console.cron-job.org/jobs>) — HTTP 204
3. CDN JSON
(<https://cdn.jsdelivr.net/gh/ChesleyLo/DownloadBankOfTaiwanExchangeRate@main/data/bot-xrt-lat-est.json>) — fetchedAtUtc (may stay old if rates unchanged)
4. Run `./scripts/trigger_update.sh` if you suspect a failure

Full guide and weekly checklist: SCHEDULING.en.md §0.5
(./SCHEDULING.en.md#05-how-to-confirm-the-schedule-ran-day-to-day)

Q4. NetSuite download fails (non-200)?

Check:

1. CDN URL spelling
2. Outbound access to cdn.jsdelivr.net
3. Deployment is Released
4. Execution Log details

Q5. Can NetSuite call the BOT site directly?

Not recommended. BOT bot-protection often blocks N/https. Always use this project's CDN JSON.

Q6. Can we use this commercially?

Rates originate from Bank of Taiwan public data. The CDN copy is a scheduled mirror and may lag. Confirm compliance internally and cite the source.

8. Document index

Document	Description
USER_GUIDE.zh-TW.md (./USER_GUIDE.zh-TW.md)	User guide (Chinese)
USER_GUIDE.en.md (./USER_GUIDE.en.md)	This user guide (English)
TECHNICAL.zh-TW.md (./TECHNICAL.zh-TW.md)	Technical guide (Chinese)
TECHNICAL.en.md (./TECHNICAL.en.md)	Technical guide (English)
../README.md (../README.md)	Project overview

To change schedules, JSON shape, or NetSuite write-back logic, see the technical guide.